

Supplementary Information

BrainEVO: A brain inspired evolving deep learning architecture for cross-domain adaptation

1 Proposition and Proof in our paper

1.1 Proposition IV.3: Gradient norm characterizes local optimization difficulty

Content of Proposition

Gradient norm characterizes local optimization difficulty. Let $z = (x, y)$ be a training sample and define the sample-wise loss $\ell_z(\Theta) = \mathcal{L}(f_\Theta(x), y)$. Assume that $\ell_z(\Theta)$ is differentiable and L -smooth with respect to Θ . For a fixed model state Θ and a step size $\eta \in (0, 1/L]$, consider one-step gradient descent on sample z :

$$\Theta_z^+ = \Theta - \eta \nabla_\Theta \ell_z(\Theta) \quad (1)$$

Then the following hold:

$$\|\Theta_z^+ - \Theta\|_F = \eta \|\nabla_\Theta \ell_z(\Theta)\|_F \quad (2)$$

and the local differences of losses can be described as,

$$\ell_z(\Theta) - \ell_z(\Theta_z^+) \geq \frac{\eta}{2} \|\nabla_\Theta \ell_z(\Theta)\|_F^2 \quad (3)$$

Therefore, under a fixed model state, a sample with a larger gradient norm requires a larger local parameter update and admits a larger guaranteed single-step loss reduction. Hence, the sample-wise learning difficulty can be quantified by any strictly increasing function of the gradient norm, i.e., $D_\Theta(z) = \phi(\|\nabla_\Theta \ell_z(\Theta)\|_F)$, where $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is strictly increasing.

Proposition IV.3 establishes that, under standard smoothness conditions, the gradient norm quantifies the local optimization effort induced by a sample, as it simultaneously determines the magnitude of the parameter update and lower-bounds the achievable one-step loss decrease. This provides a theoretical justification for using gradient magnitude as a proxy for learning difficulty.

Proof of Proposition IV.3

The full Proof of this Proposition is also provided as below.

Proof. Let $g_z(\Theta) := \nabla_\Theta \ell_z(\Theta)$ and $\Theta_z^+ = \Theta - \eta g_z(\Theta)$.

We first prove the relation between the gradient norm and the parameter update magnitude. By the definition of one-step gradient descent, $\Theta_z^+ - \Theta = -\eta g_z(\Theta)$. Taking the Frobenius norm on both sides gives,

$$\|\Theta_z^+ - \Theta\|_F = \eta \|g_z(\Theta)\|_F = \eta \|\nabla_\Theta \ell_z(\Theta)\|_F \quad (4)$$

Next, since $\ell_z(\Theta)$ is assumed to be differentiable and L -smooth with respect to Θ , by the standard smoothness inequality, for any Θ' we have

$$\ell_z(\Theta') \leq \ell_z(\Theta) + \langle \nabla_{\Theta} \ell_z(\Theta), \Theta' - \Theta \rangle + \frac{L}{2} \|\Theta' - \Theta\|_F^2 \quad (5)$$

Substituting $\Theta' = \Theta_z^+ = \Theta - \eta \nabla_{\Theta} \ell_z(\Theta)$ yields $\ell_z(\Theta_z^+) \leq \ell_z(\Theta) - \eta \|\nabla_{\Theta} \ell_z(\Theta)\|_F^2 + \frac{L\eta^2}{2} \|\nabla_{\Theta} \ell_z(\Theta)\|_F^2$. Rearranging terms, we obtain,

$$\ell_z(\Theta) - \ell_z(\Theta_z^+) \geq \left(\eta - \frac{L\eta^2}{2} \right) \|\nabla_{\Theta} \ell_z(\Theta)\|_F^2 \quad (6)$$

When $\eta \leq 1/L$, we have

$$\eta - \frac{L\eta^2}{2} \geq \eta - \frac{\eta}{2} = \frac{\eta}{2} \quad (7)$$

and therefore

$$\ell_z(\Theta) - \ell_z(\Theta_z^+) \geq \frac{\eta}{2} \|\nabla_{\Theta} \ell_z(\Theta)\|_F^2 \quad (8)$$

Hence, under a fixed model state Θ , the gradient norm of a sample determines both the magnitude of the induced parameter update and a lower bound on the achievable one-step loss decrease. This shows that the gradient norm provides a principled characterization of the local optimization difficulty of a sample. The conclusion then follows. $\square$

1.2 Proposition IV.5: Releasing regularization by decreasing p and λ facilitates parameter adaptation and increases informative knowledge assimilation from the new domain

Content of Proposition

Given the learning domain D_k and next domain D_{k+1} , we have the Dropout probability p_k and the L2 regularization coefficient λ_k . We also have the Dropout probability and L2 regularization coefficient at the next learning domain satisfying $\{p_k > p_{k+1}, \lambda_k > \lambda_{k+1}\}$. And we can denote the informativeness of knowledge encapsulated in model parameters during learning the $(k+1)$ -th domain D_{k+1} as the mutual information between learnable parameters (weights) $\mathbf{W}$ and $\mathcal{D}_{1:k+1}$, i.e., $I(\mathbf{W}_{k+1}; \mathcal{D}_{1:k+1})$. **When learning domain shift from D_k to D_{k+1} , the knowledge within model can be expanded by releasing regularization factors from learning domain k to domain $k+1$, i.e.,**

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \gamma(\mathbb{E}[I(\mathbf{W}_{k+1}; \mathcal{D}_{1:k+1})] - \mathbb{E}[I(\mathbf{W}_k; \mathcal{D}_{1:k})]) > 0 \quad (9)$$

where scaling factor $\gamma > 0$ and $\mathbb{E}[\cdot]$ denotes the expectation of such mutual information as p_k is a probability.

Proof of Proposition IV.5

Proof. As proposed in Proposition. IV.5, the dropout factor p_k is decreasing with increased k , i.e.,

$$p_k > p_{k+1} \quad (10)$$

We can divide the neurons within neural network $\mathbf{W}$ into disjointed activated neuron set $\mathbf{W}_a$ and inactivated one $\mathbf{W}_i$, satisfying $\mathbf{W} = \mathbf{W}_a \cup \mathbf{W}_i$. Given the dropout operation, we have $\forall w \in \mathbf{W}_i, H(w) = 0$. Then, the

entropy of the whole neural network can be defined as,

$$H(\mathbf{W}) = \sum_{w \in W} H(w) = \sum_{w \in \mathbf{W}_a} H(w) + \sum_{w \in \mathbf{W}_i} H(w) = \sum_{w \in \mathbf{W}_a} H(w) \quad (11)$$

The entropy of each activated neuron are equal and denoted as h_0 , then we can obtain the expectation of $H(\mathbf{W})$,

$$\mathbb{E}[H(\mathbf{W})] = \mathbb{E}\left[\sum_{w \in \mathbf{W}_a} H(w)\right] = N \times (1 - p_k) \times h_0 \quad (12)$$

where N denotes the total number of neurons in W and $\mathbb{E}[\cdot]$ denotes the mathematical expectation.

To quantify the knowledge encapsulated in the evolving $\mathbf{W}_k$ acquiring from the cumulative domains $\mathcal{D}_{1:k} = \{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_k\}$, we denote $\mathcal{M}_k$ and let it proportional to the expectation of mutual information between $\mathbf{W}_k$ and $\mathcal{D}_{1:k}$. Then we have,

$$\mathcal{M}_k \propto \mathbb{E}[I(\mathbf{W}_k; \mathcal{D}_{1:k})] = \gamma \mathbb{E}[I(\mathbf{W}_k; \mathcal{D}_{1:k})] \quad (13)$$

where $\gamma > 0$ is a positive scaling factor. In fact, the mathematical expectation of conditional entropy $H(\mathbf{W}_k | \mathcal{D}_{1:k})$ goes to decrease along with model learning process,

$$\mathbb{E}[H(\mathbf{W}_{k+1} | \mathcal{D}_{1:k+1})] < \mathbb{E}[H(\mathbf{W}_k | \mathcal{D}_{1:k})] \quad (14)$$

By replacing the mutual information with entropy computation $I(x; y) = H(x) - H(x|y)$, we can arrive that,

$$\begin{aligned} \mathbb{E}[I(\mathbf{W}_k; \mathcal{D}_{1:k})] &= \mathbb{E}[H(\mathbf{W}_k) - H(\mathbf{W}_k | \mathcal{D}_{1:k})] \\ &= \mathbb{E}[H(\mathbf{W}_k)] - \mathbb{E}[H(\mathbf{W}_k | \mathcal{D}_{1:k})] \\ &= N \times (1 - p_k) \times h_0 - \mathbb{E}[H(\mathbf{W}_k | \mathcal{D}_{1:k})] \end{aligned} \quad (15)$$

By substituting with Eq. 12 and inserting p_k, p_{k+1} , we have

$$\begin{aligned} &\mathbb{E}[I(\mathbf{W}_{k+1}; \mathcal{D}_{1:k+1})] - \mathbb{E}[I(\mathbf{W}_k; \mathcal{D}_{1:k})] \\ &= (N \times (1 - p_{k+1}) \times h_0 - \mathbb{E}[H(\mathbf{W}_{k+1} | \mathcal{D}_{1:k+1})]) \\ &\quad - (N \times (1 - p_k) \times h_0 - \mathbb{E}[H(\mathbf{W}_k | \mathcal{D}_{1:k})]) \\ &= N \times h_0 \times (p_k - p_{k+1}) \\ &\quad + (\mathbb{E}[H(\mathbf{W}_k | \mathcal{D}_{1:k})] - \mathbb{E}[H(\mathbf{W}_{k+1} | \mathcal{D}_{1:k+1})]) \end{aligned} \quad (16)$$

According to Eq. 10 and Eq. 14, we achieve,

$$\begin{aligned} &\mathbb{E}[I(\mathbf{W}_{k+1}; \mathcal{D}_{1:k+1})] - \mathbb{E}[I(\mathbf{W}_k; \mathcal{D}_{1:k})] \\ &= N \times h_0 \times (p_k - p_{k+1}) \\ &\quad + (\mathbb{E}[H(\mathbf{W}_k | \mathcal{D}_{1:k})] - \mathbb{E}[H(\mathbf{W}_{k+1} | \mathcal{D}_{1:k+1})]) > 0 \end{aligned} \quad (17)$$

Finally, we can get the following conclusion,

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \gamma(\mathbb{E}[I(\mathbf{W}_{k+1}; \mathcal{D}_{1:k+1})] - \mathbb{E}[I(\mathbf{W}_k; \mathcal{D}_{1:k})]) > 0 \quad (18)$$

Similarly, with L2 regularization factor λ_k decreases, the number of activated parameters increases, thus the same conclusion can be achieved. $\square$

2 Related work review

Learning for cross-domain adaptation. A central challenge in cross-domain adaptation lies in that distributions differ between training and testing sets [22]. Cross-domain adaptation can be divided into two aspects, the data distribution changes across space or temporal evolution for the same task, and the domain shift from source to target [57]. Early researches propose to counter temporal distribution shifts, such as traffic dynamics across years, with experience-replay of continual learning [10] and invariant learning [66] for generalization. Actually, the layout or structure of spatial domain is not always static, thus discovering the invariances on graphs including both macro networks [29] and micro biomolecules [52] also becomes a popular research branch with neuro-inspired mechanisms [35]. Specifically, urban environments are evolving on both spatial and temporal aspects [46, 60], thus dynamic explore-to-extrapolate risk objective [46] are constructed to overcome spatial shifts, while decoupled common knowledge and gradient discrepancy [24, 37, 64] are constructed to link the training and testing data with cross-domain samples. For the source-target domain shift, recent literature construct the fast-low learning systems to leverage the commonality and individual pattern during domain shifts [3, 49]. More recently, trials on cross-task learning [57, 61] establish the correlations among different tasks within similar context. Despite progress, existing methods rarely exploit shared information across inputs and quantify inter-domain relationships, so existing cross-domain adaptation lacks a unified mechanism that can be generalized into both macro and micro worlds, and enables per-domain adaptive selection within a single framework.

Evolutionary algorithm and continual learning. Evolutionary algorithms and continual deep learning are two major paradigms for adaptive models. Inspired by survival of the fittest, evolutionary computation provides a population-based meta-heuristic framework for optimization. Representative methods, such as particle swarm optimization (PSO) [15] and ant colony optimization (ACO) [25], evolve candidate solutions through stochastic search strategies resembling collective biological behaviors. However, they mainly operate under predefined optimization rules and passively explore the solution space, rather than explicitly adapting model parameters and structures to sequentially arriving data or tasks.

To enable data-adaptive evolution, surge of continual learning researches are proposed to ensure a stability-plasticity trade-off in adapting new domains or tasks with data arrival [50]. Continual learning can be summarized into four aspects. 1) Regularization-based methods [1, 23, 40] add additional terms to balance old-new data patterns, but often weaken the acquisition of new knowledge. 2) Replay-based methods exploit the gradient for experience replay [44] while generative replay replaces old samples with generated pseudo-samples, which is particularly suitable for continual zero-shot learning [16]. But they still potentially derives the mismatch between replayed and current distributions. 3) Optimization based solutions construct diverse representation or meta-learning strategies to enable model evolution [19], but lack explicit control over where to preserve and where to evolve. 4) Architecture-based methods enlarge capacity through three perspectives. **i) Progressive neural networks.** Progressive Neural Networks [42] address forgetting by allocating a new sub-network for each new task while freezing previously learned parameters. Their key advantage is strong task isolation, which effectively preserves old knowledge. However, they require continuous parameter growth as tasks accumulate and usually assume clear task boundaries to decide when a new branch should be added. **ii) Adapter-based continual learning.** Adapter-based continual learning [47] keeps a shared backbone and introduces only lightweight task-specific modules, such as adapters, prompts, or LoRA layers. This design is more parameter-efficient than progressive networks and preserves the generalization ability of the shared backbone. However, when task heterogeneity is large, a single shared backbone may be insufficient, and additional mechanisms are often needed to select or compose the appropriate adapters,

which may introduce instability [51]. **iii) Routing-based and expert-based methods.** Routing- and expert-based continual learning methods [36, 59] maintain multiple experts and use a router/gating mechanism to decide which expert(s) should process each input. Their main advantage is that they combine parameter sharing with input-dependent specialization. However, routing itself is difficult to optimize and may suffer from expert imbalance, routing collapse, or task interference, which can also result in instability of learning process. Overall, these methods indeed extend continual learning beyond fixed-capacity models. But at the same time, in the specific problem regarding domain data expansion and model, these solutions still face unresolved issues, including unbounded parameter growth, the complexity of explicit task-to-module scheduling, as well as unstable routing behavior.

In contrast, BrainEVO does not rely on adding many extra sub-networks. Instead, it performs adaptive model evolution within the existing architecture through self-organized data ordering, learnable network partition coordination with common-pattern enrichment, and LTP/LTD-inspired memory-learning interaction. This allows the model to evolve with a stable parameter scale and a more stable learning process, which is especially important when new-domain data are limited.

Learning from biological brain. Modern artificial intelligence is deeply inspired by brain structures and functions, from early neural models [41] to recent brain-inspired continual learning studies [3]. The biological brain provides important clues for adaptive intelligence from three aspects. **1) Hierarchical and inter-connected organization** [5, 20, 33]: the cerebral cortex is hierarchically organized and densely interconnected [2, 20], where lower regions process sensory details and higher regions form abstract representations, inspiring multi-level feature abstraction and inter-layer coordination in deep models [14]. **2) Complementary regional functions** [28]: the hippocampus rapidly encodes episodic experiences, while the neocortex gradually extracts stable regularities [6, 38], offering a biological basis for balancing fast adaptation and long-term memorization [31, 39, 65]. **3) Reciprocal interaction at neuronal and circuit levels:** neurotransmitter-based signaling supports synaptic strengthening and weakening, such as long-term potentiation (LTP) and long-term depression (LTD) [7, 26, 34], while circuit-level learning gradually consolidates experience into stable and generalizable knowledge [27, 28, 43, 48, 62]. Inspired by these mechanisms, episodic and working memory have been incorporated into deep models to guide plasticity and integrate new experiences into existing knowledge structures [8, 17]. Nevertheless, as highlighted by *Nature Machine Intelligence* [9], the integration of AI and neuroscience remains largely open. In cross-domain adaptation, key challenges still include how to operationalize easy-to-hard curricula and how to mimic brain-like learning progress and knowledge interaction for reliable pattern transfer [12].

3 Additional experiments

3.1 Detailed analysis on gradient-based task difficulty measurement and curriculum learning process

In this subsection, we provide more detailed analysis on evaluating the effectiveness, and efficiency gradient-based task difficulty measurement.

Visualized extracted gradient for task difficulty quantification

To help better understand the gradient flow patterns, we not only visualize the extracted gradient to measure the task difficulty, but also provides how many domains in each dataset and the learning epochs experience

for each task difficulty measurement in Fig. 1. To ensure the quality of visualization, we truncate the dimension of gradient vectors via the minimized one and the visualized results offer the informative abstract for potential data-model coupling patterns.

The number of truncated training epoch is 3 for pre-heating and capturing data patterns on spatiotemporal datasets, while such epoch is 5 for molecule datasets. The differences between two types of datasets may lie in that 1) the diversity and complexity of functional groups in molecules calls for more sufficient pattern extraction, while spatio-temporal datasets exhibit strong periodicity and repetitive dynamics, 2) ST data learning are with larger while the scale molecule datasets are relatively, and such training epochs ensure the the efficiency and provide tradeoff of time costs during training. And the gradient maps demonstrate that these gradients are not near-zero vectors, but with distinguished and informative patterns for extracting data regularity.

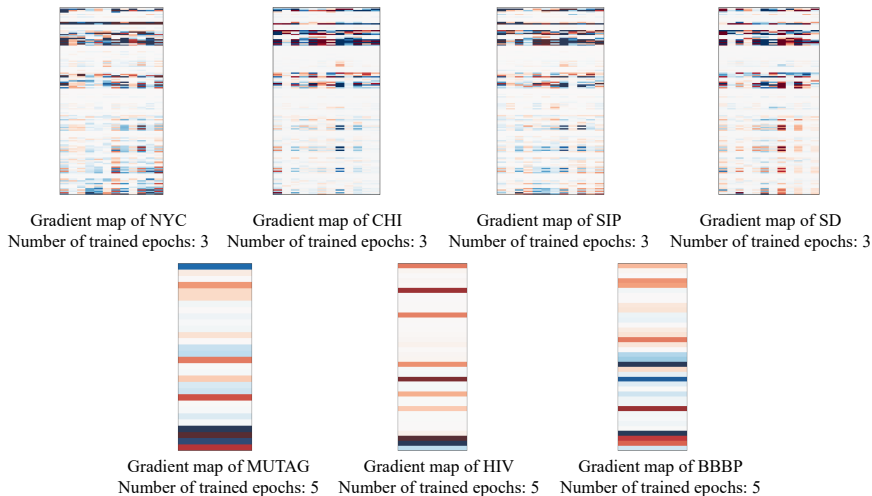


Figure 1: Training epochs and gradient maps at learning stability.

Complexity and efficiency of curriculum learning process

Our hierarchical curriculum learning provides a data organization strategy to deal with disordered data, which is a one-off domain-level pre-processing rather than a repeated cost during every update. To offer further evaluation, we first show the complexity of curriculum learning and then provide the evidence of tradeoff between effectiveness and efficiency, regarding performances, number of learning samples, as well as dimensions of gradient vectors.

First, **the runtime ratios (time cost proportion)** between implementing curriculum learning and full learning process for each dataset are shown in Tab. 1, which illustrates the number of divided domains for reference. The results explicitly show that the proportion of curriculum learning is acceptable to overall training process.

Second, to facilitate the process of learning difficulty quantification, we can borrow the idea of stochastic gradient, and construct **a lightweight probe-based curriculum variant**, by either estimating the early-stage gradients under a small training budget instead of full convergence, or compressing the gradient vectors into reduced ones (i.e., reduce the dimensions of gradient vectors). In this way, we can randomly take 40%, 50%, 60%, 70% samples to take part in gradient computation, or compress the gradient vectors into

Table 1: The runtime statistics for curriculum learning

Dataset	NYC	CHI	SIP	SD	MUTAG	HIV	BBBP
Number of divided domains	16	12	8	6	4	4	4
Runtime ratio of CL (%)	10.81	10.65	9.75	9.34	17.58	12.69	16.82

Table 2: Performance evaluation with reduced sample-level computational complexity on NYC

Prop. of training samples	40%	50%	60%	70%	100%
Runtime ratio of CL (%)	4.92	5.76	6.74	7.99	10.81
Performance (MAE)	6.559	6.555	6.532	6.503	6.496
Prop. of gradient vector dimension	40%	50%	60%	70%	100%
Runtime ratio of CL (%)	4.86	5.91	6.68	7.92	10.81
Performance (MAE)	6.558	6.548	6.542	6.499	6.496

40%, 50%, 60%, 70% of previous ones for clustering, thus reducing the overall computational complexity. Here we present the performances vs computational burden (runtime ratio, %) on NYC for verifying such effectiveness-efficiency tradeoff in Tab. 2.

Result summary. According to above results, on the one hand, it is demonstrated the overall training time proportions of different datasets are ranging from 9.34% to 16.82%, which are not heavily on the whole process. On the other hand, when either reduce the learnable samples or reduce the dimension of gradient vectors, the results do not go downward heavily and show the potential of tradeoff on effectiveness-efficiency. Furthermore, the ratio is also determined by the scale of dataset and complexity of backbone network.

Broader discussion on gradient for curriculum learning and task difficulty quantification

As measuring the difficulty degree of learning a series data is important. We here offer some more insights and discussions on our gradient-based curriculum learning.

Gradient-based difficulty estimation may be affected by randomness from initialization, batch order, and training stage, making single-run gradients an unstable proxy for curriculum construction. A practical solution is to estimate difficulty through *repeated stochastic observations* rather than a single gradient snapshot. Specifically, for each sample or domain, gradients can be collected across multiple random training runs and several early-stage checkpoints, and then summarized by a robust statistic. To further reduce scale inconsistency across runs, we suggest aggregating *ranks* instead of directly averaging raw gradient magnitudes. In this way, the difficulty of a sample/domain is interpreted as whether it consistently induces larger optimization pressure across training trajectories, rather than whether it happens to produce a large gradient in one run. Such a strategy can further improve the stability and reliability of curriculum learning under stochastic optimization.

3.2 Baseline, evaluation settings and results

In this section, we elaborate the settings of baselines and introduce one more backbones for both research fields of urban science and biological science, to further demonstrate the fairness of evaluation process, superiority of learning strategy and scalability of BrainEVO.

We exploit comprehensive baselines with various training strategies to evaluate the effectiveness of our BrainEVO.

Baselines and metrics for Spatiotemporal urban datasets. We employ three standard performance metrics, i.e., Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). Furthermore, our study incorporates 11 baseline models for comparative assessment including 6 STGNNs with different backbones and learning manners, an RNN-based model, 3 attention-based models and a NeuroAI model. We especially take GWN and CMuST as backbones for BrainEVO on spatiotemporal datasets and take GIN and GAT for BrainEVO on molecule datasets.

- *STGCN*: a spatial-temporal aware graph learning method for POI recommendation [58] (STGNN).
- *STGODE*: Spatial-Temporal Graph ODE Networks for Traffic Flow Forecasting [13] (STGNN).
- *GWN*: A graph-based traffic prediction model that integrates TCNs and GCNs [53] (STGNN).
- *GWN-FR*: Train GWN in a full randomly sampling manner [53] (STGNN).
- *GWN-w/o Com*: Train GWN directly on testing dataset [53] (STGNN).
- *STGP*: A prompt-based framework capable of adapting to diverse tasks in a data-scarce domain [21] (STGNN and multi-source based).
- *AGCRN*: Adaptive graph convolution recurrent network for traffic forecasting [4] (RNN-based model).
- *STTN*: Spatial-temporal transformer networks for traffic flow forecasting [55] (Attention-based model).
- *ASTGCN*: Attention based spatial-temporal graph convolution networks for traffic flow forecasting [18] (Attention-based model).
- *CMuST*: A continuous multi-task spatio-temporal learning frame-work [57] (Attention-based model and multi-source model).
- *SynEVO*: A neuro-inspired spatiotemporal framework for cross-domain adaptation [32] (NeuroAI model).

Baselines and metrics for biological molecular datasets. For these molecule property modeling, they are mostly classification tasks and we employ three standard metrics, Accuracy, ROC-AUC and F1-score. And we incorporates six baseline models for comparative assessment.

- *GAT*: A graph neural network model that employs attention mechanisms to aggregate information from neighboring nodes [45].
- *GCN*: A graph neural network model that performs neighborhood aggregation through spectral-based graph convolutions [63].
- *GIN*: A graph neural model that uses injective aggregation to achieve powerful graph representation and discrimination [54].
- *GIN-FR*: Train GIN with full randomly sample orders [54].
- *GIN-w/o Com*: Directly train GIN on targeting samples without considering cross-domain commonality [54].

- *PNA*: A graph neural model that integrates multi-aggregator and degree-scaled mechanisms for expressive neighborhood representation [11].
- *EMoNet*: An environment causal learning for molecule ood generalization [30].
- *NeGo*: A negative inference graph OOD framework [56].

Implementation details. To facilitate the reproducibility of BrainEVO, we provide more details of training and implementations. The datasets were partitioned into training, validation, and testing sets with a ratio of 7:1:2. We follow the general continual learning protocol [50] and allow source domains to be trained and leave target domains as evaluation domains for inference, where targeted samples cannot be accessible during training process. According to the characteristics of datasets themselves, we take the NYC, CHI, and SIP datasets for both cross-source and cross-temporal domain validation, and take SD set for cross-temporal domain evaluation due to its large time span with single attribute.

Learning across domains on urban datasets. For cross-source domain evaluation on the first three spatiotemporal datasets (NYC, CHI, SIP), we respectively reserve TaxiPick, TaxiDrop, and Speed as the target evaluation domains within their datasets, and iteratively training on the remaining data sources. For cross-temporal evaluation on these datasets, each day was divided into four equal intervals, with the final intervals across all source domains held out for assessment. For cross-temporal evaluation on SD dataset, each day was segmented into six equal periods, and the last period was reserved to evaluate temporal domain adaptation.

Learning across domains on molecule datasets. Biological molecules consist of invariant structures like scaffolds and pharmacophores, as well as environment sensitive elements where different substructures imply various properties. To this end, we partition molecules into four domains by whether the molecule has carbonyl group or not and whether it has halogen or not, as these functionalities induce systematic shifts in physicochemical properties, i.e., halogens enhance lipophilicity and stability, while carbonyl groups are highly polar and prone to nucleophilic addition. This yields chemically meaningful domain differences, and learning across these specific domains verifies whether our evolution learning can separate invariant representations from environment-dependent mechanisms. Considering that those molecules with neither carbonyl group nor halogen are more difficult to predict the chemical properties because of lacking the important mentioned groups, we leave the domain which contains molecules with neither carbonyl group nor halogen as the target evaluation domain and utilize other domains as training domains.

Result comparisons.

Comparisons on urban spatiotemporal datasets. The comparison results are reported in Table. 3, which presents the performance of cross-source adaptation on the NYC, CHI, and SIP datasets, as well as cross-temporal adaptation on the SD dataset. In general, our BrainEVO consistently outperforms all baselines across most metrics, achieving either the best or the second-best performance on all four datasets. The improvement beyond best baselines can be found in the last line of Table. 3. Compared with existing baselines on cross-source adaptation, BrainEVO achieves the best overall results. To be specific, BrainEVO reduces prediction errors by averagely 18.8% on NYC, 4.2% on CHI, 7.3% on SIP against all baselines in terms of MAE, which confirms the generalization performance of our method. It’s worth noting that BrainEVO reduces MAE by about 24.9% on NYC, 4.7% on CHI, and 8.1% on SIP compared to non-evolving baselines, which confirms the significance of evolving capacity of deep model to deal with domain adaptation. On the cross-temporal dataset SD, BrainEVO sets a new state of the art as well, improving upon baselines by 16.5% on MAE. This demonstrates that BrainEVO can robustly capture temporal dependencies and maintain

Table 3: Performance comparison on cross-source adaptation on NYC, CHI, SIP and cross-temporal adaptation on SD. Best results are **bold** and the second best results are underlined.

Methods	NYC			CHI			SIP			SD		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
STGCN	6.774	15.853	0.374	1.518	2.804	0.415	0.753	1.492	0.219	12.477	22.055	0.182
STGODE	9.522	22.555	0.481	1.543	2.792	0.433	0.732	1.460	0.211	12.300	20.808	0.180
GWN	10.263	24.535	0.546	1.520	2.873	0.421	0.737	1.473	0.212	18.890	29.220	0.230
GWN-FR	10.076	24.128	0.537	1.522	2.870	0.419	0.738	1.472	0.212	17.661	28.987	0.226
GWN-w/o Com	8.963	21.435	0.489	1.603	2.918	0.421	0.747	1.484	0.217	14.792	25.583	0.190
STTN	7.962	19.544	0.435	1.494	2.699	0.409	0.732	1.461	0.212	13.092	22.054	0.183
AGCRN	8.254	19.301	0.488	1.543	2.805	0.427	0.743	1.469	0.215	12.225	22.094	0.179
ASTGCN	10.323	25.070	0.519	1.536	2.809	0.413	0.743	1.473	0.215	13.079	22.047	0.185
CMuST	6.576	14.954	0.459	1.498	2.781	0.388	0.737	1.497	0.219	10.940	19.113	0.162
SynEVO	<u>6.494</u>	14.885	0.358	<u>1.486</u>	2.733	<u>0.361</u>	<u>0.697</u>	1.390	<u>0.205</u>	10.984	18.654	0.157
STGP	6.497	14.882	0.373	1.482	2.678	0.364	0.695	1.392	0.202	17.906	28.651	0.188
BrainEVO (GWN)	6.496	<u>14.422</u>	0.367	1.453	2.682	0.364	<u>0.680</u>	<u>1.350</u>	<u>0.198</u>	<u>10.585</u>	<u>17.839</u>	<u>0.143</u>
BrainEVO (CMuST)	6.492	14.421	<u>0.362</u>	1.373	2.484	0.350	0.661	1.321	0.196	9.467	17.528	0.141

Table 4: Performance comparison on cross-domain adaptation on biological molecular datasets. Best results are **bold** and the second best results are underlined.

Methods	MUTAG			HIV			BBBP		
	ACC(%)	AUC(%)	F1-score(%)	ACC(%)	AUC(%)	F1-score(%)	ACC(%)	AUC(%)	F1-score(%)
GAT	76.4 ± 4.7	79.2 ± 3.1	76.1 ± 4.3	70.9 ± 2.3	72.5 ± 3.0	79.9 ± 4.1	65.4 ± 2.6	69.8 ± 3.4	72.9 ± 3.3
GCN	83.6 ± 5.3	87.6 ± 5.4	82.0 ± 3.7	75.1 ± 1.7	76.3 ± 2.1	82.5 ± 3.7	65.0 ± 2.1	68.7 ± 2.7	70.3 ± 3.6
GIN	89.3 ± 5.2	90.2 ± 4.9	86.6 ± 3.9	75.6 ± 1.4	77.2 ± 1.2	82.6 ± 2.3	66.4 ± 1.9	70.1 ± 2.1	71.7 ± 2.6
GIN-FR	88.6 ± 4.9	90.4 ± 4.7	86.9 ± 4.1	75.8 ± 1.5	77.3 ± 1.6	82.5 ± 2.1	66.9 ± 1.8	70.0 ± 1.7	71.7 ± 2.8
GIN-w/o Com	92.9 ± 3.1	95.7 ± 2.2	90.5 ± 1.3	77.3 ± 1.1	79.9 ± 1.5	84.6 ± 1.4	71.2 ± 1.6	74.9 ± 1.0	77.0 ± 1.1
PNA	86.4 ± 4.2	88.4 ± 3.5	82.4 ± 4.0	77.5 ± 1.8	80.1 ± 1.1	84.1 ± 0.9	71.7 ± 1.9	74.7 ± 1.7	76.8 ± 1.7
EMoNet	96.4 ± 1.5	96.1 ± 2.3	91.2 ± 1.4	77.8 ± 1.6	80.3 ± 1.5	84.9 ± 1.1	72.1 ± 1.1	75.8 ± 0.8	77.2 ± 1.3
NeGo	92.9 ± 1.7	95.5 ± 1.1	90.1 ± 1.6	76.4 ± 1.4	79.7 ± 0.9	84.2 ± 2.2	70.9 ± 0.9	73.9 ± 1.2	76.5 ± 1.1
BrainEVO(GIN)	97.9 ± 1.3	99.3 ± 0.9	91.5 ± 1.2	81.5 ± 1.0	82.3 ± 1.6	86.3 ± 1.7	75.1 ± 1.7	78.3 ± 1.3	79.0 ± 1.6
BrainEVO(GAT)	88.6 ± 1.5	93.8 ± 1.0	86.7 ± 1.9	76.8 ± 1.0	77.2 ± 1.5	82.4 ± 1.9	73.9 ± 2.0	77.6 ± 1.5	79.8 ± 1.5

stable generalization under non-stationary time dynamics. We also altered the training strategy, including using a fully random sample order and training only on the target domain, to validate the importance of our proposed sequential learning and common-pattern learning mechanisms. And it should be emphasized that BrainEVO consistently improve the performances against SynEVO, confirming the effectiveness on domain transfer when incorporating hierarchical learning and learning-memory interaction mechanism.

Besides, concentrating on the comparison with backbone GWN, BrainEVO significantly outperforms GWN by large margins (about 36.7% on NYC, 4.4% on CHI, 7.7% on SIP and 44.0% on SD in terms of MAE), further verifying that the adaptive evolving scheme has an edge on helping the backbone model jump out of local optimal, which boosts transferability and generalization ability in challenging cross-domain spatiotemporal prediction tasks.

Moreover, by incorporating our BrainEVO with two respective backbones, GWN and CMuST for spatiotemporal datasets while GIN and GAT for biological molecule datasets, we can see that our BrainEVO can seamlessly couple with other spatiotemporal learning architecture and graph-based learning module, and achieve competitive performances. In addition, BrainEVO+CMuST outperforms BrainEVO+GWN but sacrifices more computational costs [32]. Overall, based on above results, we can conclude that our backbone coupling solution still makes sense with both effectiveness and efficiency, and remain these couplings in our main text.

Comparisons on biological molecular datasets The comparison results on micro biological molecular datasets are reported in Table. 4, illustrating performance of functional group-level cross-domain adaptation on MUTAG, HIV and BBBP datasets. Overall, BrainEVO achieves the best accuracy on domain transfer

Table 5: Learning performances without cross-source out-of-distribution setting (Accuracy reported)

Dataset	MUTAG	HIV	BBBP
GAT (i.i.d.)	0.836	0.779	0.768
GIN (i.i.d.)	0.929	0.896	0.831
BrainEVO (i.i.d.)	0.979	0.925	0.875
BrainEVO (Averaged OOD)	0.980	0.818	0.753

tasks regarding all datasets, demonstrating consistent advantages over existing graph-based baselines. In particular, BrainEVO improves accuracy by about on average 12.5% on MUTAG, 8.1% on HIV, 9.7% on BBBP compared to all baselines, especially outperforms second best EMoNet by about 1.6% on MUTAG, 4.8% on HIV and 4.2% on BBBP, which confirmed the adaptation capability of our method for dealing with biochemical distribution shifts. These results demonstrate that BrainEVO can effectively transfer common patterns between molecules, while maintaining robustness across diverse datasets. Moreover, when comparing with backbone GIN, BrainEVO outperforms GIN by large margins, i.e., about 10.2% on average across all biological molecular datasets in terms of ACC. And we also alter the backbone to GAT, and find that BrainEVO+GAT outperforms most baselines but inferior to BrainEVO+GIN. Thus BrainEVO+GIN becomes the gold combination and all these results can further demonstrate that BrainEVO improves transferability and generalization in cross-domain molecular property prediction tasks.

3.3 Detailed analysis of degeneration on molecule tasks

To further verify that distribution shift indeed causes severe degradation, we additionally report results under a standard random train-test split, where all sample components are mixed and 30% of the data are randomly held out for testing, without using our composition-based grouping, i.e., training and testing on the same domain. The results are demonstrated in Tab. 5. We especially take the averaged OOD results (averaging the performances of four settings "++/+-/-+/-" of original manuscript Tab.V) as the comparison on the last line of Tab. 5. Under this conventional setting, the performance on HIV and BBBP is much higher than our deliberately constructed OOD settings (92.5% vs 81.8% on HIV and 87.5% vs 75.3%). This confirms that the severe degradation is real and is caused by the simulated distribution shift induced by our sample-group partition protocol.

Therefore, the relatively stable performance of some baselines across the ++ and -- subsets does not imply that the domain shift is weak. Rather, it indicates that all compared methods are being evaluated under the same challenging non-i.i.d. protocol, while the additional comparison with standard random splitting confirms that the constructed setting is indeed much harder and leads to clear performance degeneration.

3.4 Details of ablation study

For ablation study, we have added one metric, Average Improved Performance (AIP), to measure how overall model improves the performance against the module-level ablative variant by computing the averaged decreases on MAPE (for regression tasks), and averaged increases on ACC (for classification tasks) across all datasets, i.e.,

$$AIP = \frac{1}{N} \left(\frac{MAPE_V - MAPE_{BrainEVO}}{MAPE_V} + \frac{ACC_{BrainEVO} - ACC_V}{ACC_V} \right) \quad (19)$$

Table 6: Ablation studies of cross-temporal tasks on spatiotemporal sets and cross-domain adaptation on biological molecular sets.

Methods	NYC			CHI			SIP			SD			MUTAG		HIV	BBBP	Improvement (AIP) $\pm$ std
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	ACC	ACC	ACC	ACC	/
BrainEVO	6.308	14.289	0.365	1.451	2.657	0.363	0.672	1.338	0.191	10.585	17.839	0.143	0.979	0.815	0.751	/	
BrainEVO-HCL (intra-cluster)	7.007	16.661	0.374	1.696	3.239	0.380	0.699	1.389	0.204	12.071	19.932	0.183	0.957	0.799	0.742	5.80% $\pm$ 0.067	
BrainEVO-HCL (inter-cluster)	6.763	15.982	0.367	1.667	3.146	0.370	0.691	1.376	0.203	11.962	19.334	0.177	0.964	0.804	0.741	4.54% $\pm$ 0.062	
BrainEVO-Ela (static)	7.639	17.315	0.396	1.713	3.179	0.449	0.723	1.421	0.212	13.473	20.880	0.226	0.943	0.793	0.737	11.73% $\pm$ 0.11	
BrainEVO-Ela (inverse proportion)	6.501	14.763	0.367	1.474	2.719	0.366	0.680	1.353	0.196	10.994	18.045	0.148	0.971	0.807	0.743	14.55% $\pm$ 0.009	
BrainEVO-Ela (linear proportion)	6.577	14.952	0.368	1.488	2.801	0.368	0.684	1.369	0.199	11.046	18.243	0.150	0.971	0.806	0.744	19.60% $\pm$ 0.015	
BrainEVO-MIL ($\rho = 0$)	6.626	15.005	0.384	1.641	2.846	0.392	0.720	1.378	0.215	13.123	21.285	0.179	0.921	0.787	0.729	8.07% $\pm$ 0.055	
BrainEVO-MIL ($\kappa \rightarrow +\infty$)	7.104	16.719	0.376	1.702	3.331	0.379	0.717	1.365	0.214	13.207	21.772	0.183	0.914	0.784	0.72	7.71% $\pm$ 0.063	
BrainEVO-LCM	6.695	15.207	0.385	1.569	2.851	0.372	0.674	1.341	0.197	11.128	18.645	0.165	0.964	0.798	0.746	4.05% $\pm$ 0.04	

Table 7: Detailed evaluations on LCM-based ablative variant on NYC and BBBP (MAE Performance/Runtime ratio)

Evaluation details	Full training set	With 67% training set	With 33% training set
	(Performance/Runtime ratio)	(Performance/Runtime ratio)	(Performance/Runtime ratio)
NYC, BrainEVO with LCM	6.308/0.38%	6.704/0.41%	7.439/0.42%
NYC, BrainEVO-w/o LCM	6.695/NULL	7.171/NULL	7.802/NULL
BBBP, BrainEVO with LCM	0.751/0.44%	0.729/0.46%	0.707/0.50%
BBBP, BrainEVO-w/o LCM	0.746/NULL	0.721/NULL	0.685/NULL

The variances of performance decreases are also incorporated into this metric to illustrate the robustness. This additional analysis provides a clearer and more direct illustration of the individual contribution and stability of each component, rather than relying only on the final performance numbers in the original tables. The added AIP pm std performance evaluations have been added into the most right column of Tab. 6.

From Tab. 6, we can easily decouple the importance of each module. First, the elastic module, i.e., dynamic adjustment of dropout factor and weight decay plays most important role and then memory-learning mutual interaction and finally the hierarchical curriculum learning.

3.5 Detailed evaluation of Learning Construct Memory (LCM)

Monitoring the activation frequencies of all neurons may cause many external computational costs. Actually, in our LCM, we would like to clarify that it does not monitor neuron activation frequencies in a layer-by-layer manner across the entire network. Instead, in our implementation, we only compute activation-frequency statistics for the *last neural activation layer*, since this layer is the most directly related to the final prediction output. We believe this strategy can well balance the effectiveness and efficiency. Therefore, the additional computation is limited where it only involves a lightweight vector-level aggregation over the existing activation tensor, without any extra backward passes or additional optimizer states.

To make this point more precise, we report the actual runtime with and without (-w/o LCM) on NYC and BBBP, where the running time of the full model against its ablated variants (-w/o LCM). The results have been demonstrated in Tab. 7. As illustrated, with LCM module, BrainEVO is more capable of countering data deficiency challenge, where the errors on NYC keeps much lower and accuracy on BBBP keeps higher as sparsity rate increasing. On computational cost aspect, our BrainEVO absolutely deserves LCM module as only approximately 0.5% overall running time can exchange the valuable robustness under extreme scenarios.

Table 8: Evaluation (MAE) of ablative variant of BrainEVO-LCM under catastrophic forgetting setting

Models	NYC (Cross-source) (TaxiDrop/Crowd-in/Crowd-out)	NYC (Cross-temporal) Test on former three divisions	BBBP (Cross-source) Train on ++ then -- and test on ++
BrainEVO with LCM	11.950/20.914/31.088	19.749/37.939/39.777	0.700
BrainEVO-w/o LCM	12.192/21.232/32.520	19.858/37.957/41.074	0.686
Backbone	15.847/27.939/40.649 (GWN)	23.923/42.659/46.747 (GWN)	0.626 (GIN)

Table 9: Evaluation on forgetting issue (MAE reported)

Models	NYC (Cross-source) (TaxiDrop/Crowd-in/Crowd-out)	NYC (Cross-temporal) Test on former three divisions	Models	BBBP (Cross-source) Train on ++ then -- and test on ++
GWN	15.847/27.939/40.649	23.923/42.659/46.747	GIN	0.626
CMuST	12.355/22.057/34.123	20.125/38.141/44.764	GAT	0.575
BrainEVO	11.950/20.914/31.088	19.749/37.939/39.777	BrainEVO	0.700

3.6 Detailed evaluation of BrainEVO’s forgetting issue

To provide comprehensive evaluation on forgetting issue of our BrainEVO, we design two respective experiments.

The overall performance when testing on call-back datasets. Concretely, for evaluating the forgetting issue, we let the model call back to re-test on previous learned source domain when it has finished fine-tuning on last targeted domain. The detailed evaluation protocols are as follows.

- On NYC cross-source setting, BrainEVO has just fine-tuned on targeted TaxiPick, then it is required to make predictions on 10% of total samples on previous three domains, TaxiDrop, Crowd-in and Crowd-out, following 7:2:1 sample division paradigm.
- On cross-temporal setting, we have divided the training set into four division, [0-6 hour, 6-12 hour, 12-18 hour, 18-24 hour], and we lastly train on 18-24h and then retrieve the previous three divisions with 10% sampling rate for evaluating the forgetting issue.
- On molecule dataset, we let BrainEVO sequentially trains on ++, and --, then we allow the testing on previous ++, where the sampling rate also keeps 10%.

Following above protocol, we report the performances of MAE metric in Tab. 9 for your convenience. As observed, we can see that our BrainEVO still performs robust against peer models when competing on the scenarios of catastrophic forgetting.

Evaluations with ablative LCM module under forgetting scenarios. We further report a re-visiting evaluation on previously learned domains in Tab. 8, comparing the full model against the variant without LCM, in order to directly assess the contribution of LCM to mitigating forgetting. We can see that performances of ablative variants without LCM decrease ranging from 1.25% to 2.63%. In addition, our solution does not work in an independent manner, but it improves the robustness and alleviates forgetting in a jointed perspective with controllable elastic expansion, where the prominent performance decrease can be observed when comparing against Backbones (GWN on NYC and GIN on BBBP).

3.7 Detailed hyperparameter evaluation of BrainEVO

The major hyperparameters in BrainEVO can be categorized into four aspects.

- Dropout p_0 from $\{0.1, 0.3, 0.5, 0.7, 1\}$, controlling the scale of activated neurons for elastic model evolution.
- Weight decay factor of L2 regularization, λ_0 from $\{0.01, 0.03, 0.05, 0.07, 0.1\}$, which adjusts the model complexity.
- Distance threshold κ from $\{1, 2, 3, 4, 5\}$, denotes the minimum distance across two domains.
- The low and high frequency threshold of neuron activations, $\epsilon_1 \in \{0.1, 0.2, 0.3, 0.4\}$ and high frequency $\epsilon_2 \in \{0.6, 0.7, 0.8, 0.9\}$ vary across two respective datasets on urban (NYC) and molecule (BBBP) sets.

The results can be found in Fig. 2-Fig. 5.

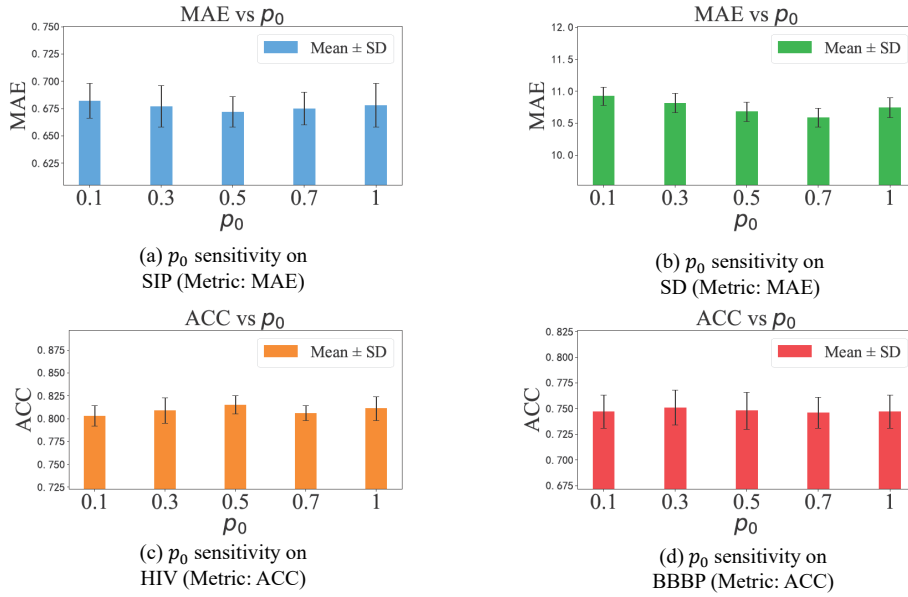


Figure 2: Updated parameter sensitivity analysis of initialized Dropout probability p_0 .

Analysis on Dropout p_0 . As observed, the performance variation patterns are similar across spatiotemporal datasets and molecule datasets (tasks), as ST datasets arrive best around $p_0 = 0.5$ while molecule datasets also behave in this way. This demonstrates 50% can be a general good probability for neuron activation at an initial stage, benefiting following evolving process.

Analysis on weight decay initialization λ_0 . The λ_0 arrives best at 0.03 on spatiotemporal datasets, while 0.03 or 0.05 for molecules, where it reveals the consistency of general neural network behaviors in fitting diverse datasets.

Analysis on domain distance threshold κ . The parameter κ serves as a critical threshold governing the incorporation of new domains. An excessively small κ prohibits the admission of any new domain, thereby contradicting the principle of ‘harmony with diversity for collective intelligence’. Conversely, an overly large κ admits noisy or irrelevant domains under overly relaxed fusion conditions, which ultimately degrades overall performance. Therefore, an appropriate value of κ must be carefully selected to strike a balance between extracting commonality and preserving distinctive domain-specific features, ensuring both robustness and adaptability. Specifically, we observe that these four datasets consistently arrive $\kappa = 2$ for the best, which may indicate the similar model behaviors across different tasks.

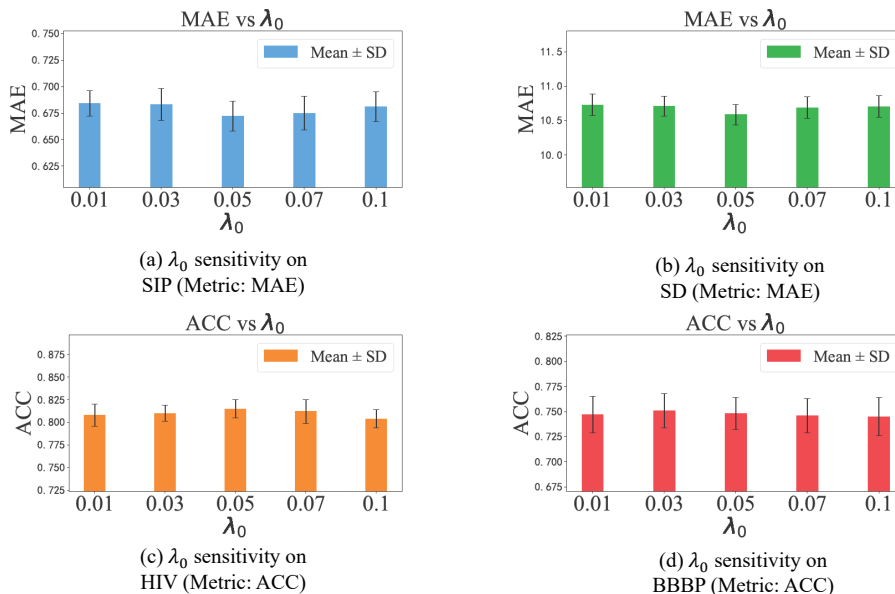


Figure 3: Updated parameter sensitivity analysis of initialized weight decay λ_0 .

Analysis on memory band (ϵ_1, ϵ_2). The ϵ_1 defines the low frequency of neuron activation while while ϵ_2 defines the high-frequency activeness and stability of neurons during learning process. For spatiotemporal datasets, we selected the hyperparameters yielding lower MAE, while for molecular property datasets, we selected those yielding higher Accuracy. These hyperparameters essentially characterize the model’s tolerance range for moderately activated patterns. Due to differences in pattern stability and gradient consistency across data domains, neural networks may form memory selection bands of different widths, which is conceptually consistent with the biological LTP/LTD mechanism.

Specifically, spatiotemporal data usually exhibit stronger periodicity and continuity, leading to more stable patterns, more clustered embeddings, and more consistent gradient directions. As a result, the activation distribution is more concentrated, so the suppression boundary does not need to be very low and the enhancement boundary does not need to be very high, resulting in a relatively compact interval. In contrast, molecular datasets involve more diverse functional groups and combinatorial structures, leading to higher pattern diversity and weaker gradient consistency. Therefore, molecular tasks are closer to sparse causal pattern discovery, where only patterns that are highly stable, repeatedly observed across samples, and consistently beneficial to loss optimization are reinforced.

4 Reproducibility

All the resources, including datasets, code of models and evaluations as well as this Supplementary Information associated with our manuscript are available at <https://github.com/Rodger-Lau/BrainEVO>.

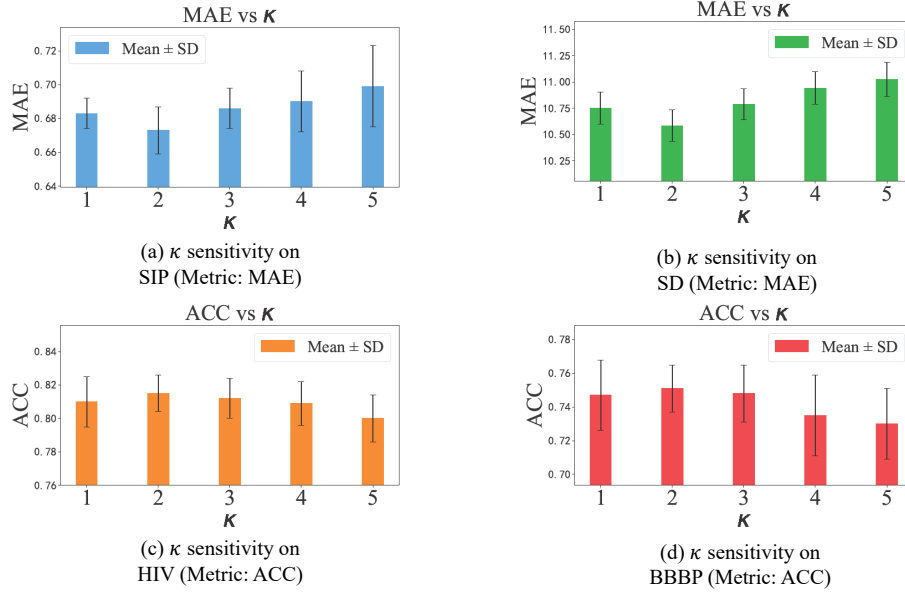


Figure 4: Updated parameter sensitivity analysis of domain distance threshold κ .

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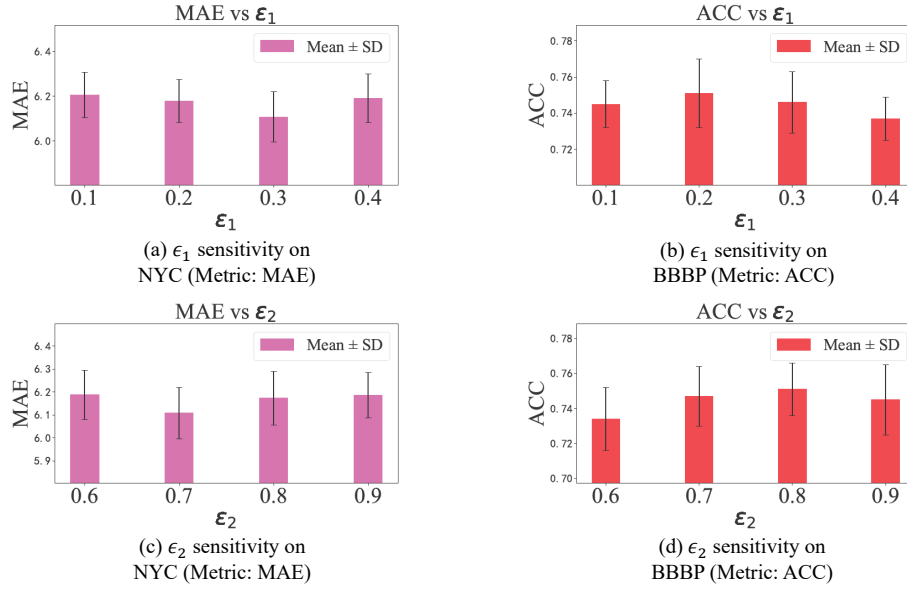


Figure 5: Updated parameter sensitivity analysis of ϵ .

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